

Software Planning and Review Process

Summary

Planning and reviews are often not applied by software project managers, yet these processes— when mastered— will have the greatest paybacks for software organizations. Organizations are so eager for quick, tangible results that the planning and review process is not applied at the correct level of detail and thoroughness. The lack of adequate plans and reviews severely reduces project capability to meet budget, schedule, and technical commitments. The results include cost overruns, missed schedules, and customer dissatisfaction.

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Software Planning and Review Process

Many software projects fail because project management is inadequate and managers are lax in adherence to the appropriate project management processes. Software project managers often do *not* apply the five basic processes of project management in a disciplined consistent manner. These five processes are:

1. Project initiation and scope definition
2. Planning
3. Execution and control
4. Review and evaluation
5. Closure

Of the five processes, planning and reviews/evaluation are the processes that— when mastered— have the greatest paybacks for software organizations. Most managers believe that managing a software project is a unique experience without commonalities to other types of project management jobs. This is only partially true. Software projects do have some unique characteristics that make them difficult to manage. The seven major characteristics (described by Watts Humphrey, *Introduction to Software Process Improvement*, 1992) that make software projects difficult to manage include:

- Software is generally more complex.
- Software changes appear relatively easy to make.
- Many late-discovered hardware problems are addressed through software changes.
- Because of its low reproduction cost, software does not have the natural discipline of release to manufacturing.
- Software discipline is not grounded in natural science, and it lacks ready techniques for feasibility testing and design modeling.
- Software is often the element integrating an entire system, thus adding to its complexity and creating exposures to late change.
- Software is generally most visible, thus most exposed to requirements changes and most subject to user complaint.

The major causes of management failure, however, are not that software projects are unique in managing or have unique characteristics. Failure occurs when the processes of project management are not applied in a disciplined consistent manner.

The Planning Process

Big Picture Planning Phase

The most important management process in any complex software development project is the planning process. The plan should define assignment responsibility and quality criteria. The primary benefit of a plan is that the project has established a "base line" for the scope, risks, and resources required to do a process or task. Another benefit is that each plan provides information to all personnel on the procedures required approaching a task, and the plan is an excellent communication tool for informing all project members about the "big picture." The plan becomes the map for how the task or process will be accomplished in relationship to the other project plans.

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Software development project plans allow knowledgeable people to define, debate, and modify a planned action in advance of producing software code. This is the phase of the project when changes can take place with minimal resource impact. All plans should be prepared in enough detail prior to the final project go-ahead. The plan should present a clear definition of the purpose of the project, including the costs, schedules, and risks involved. The basic input used for the project plans is the high-level data produced during the concept and definition phase. The information is:

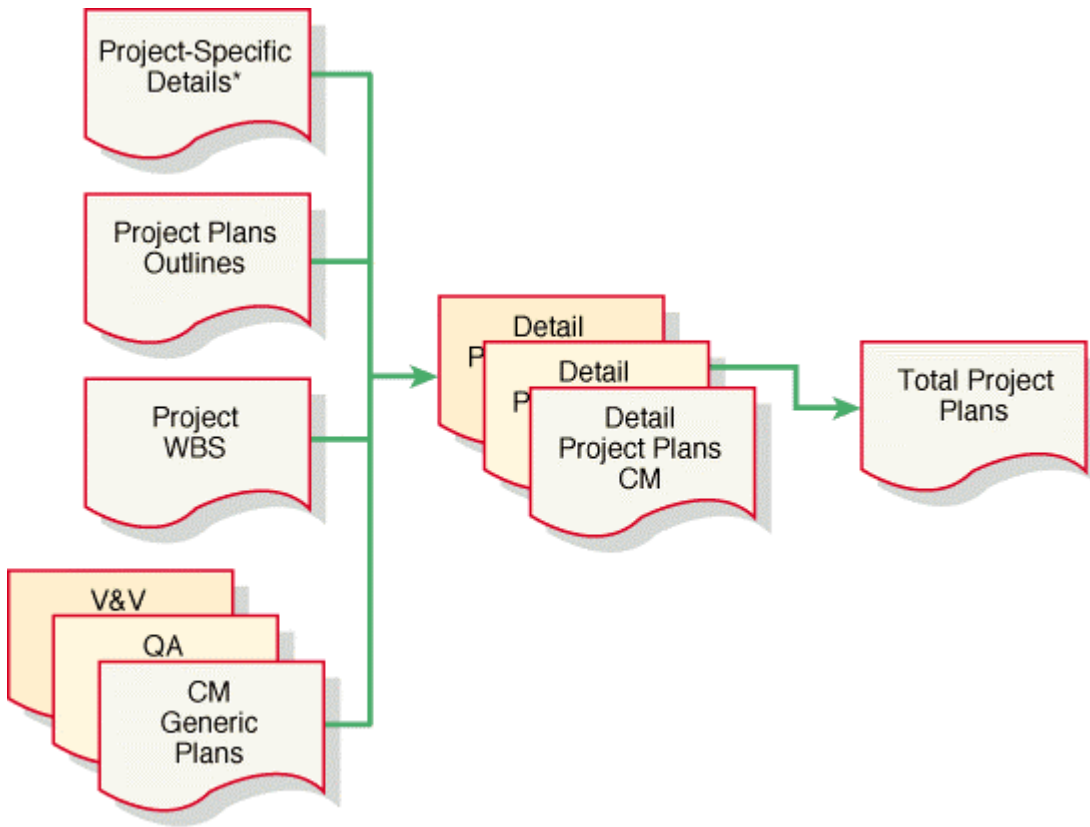
1. Project-specific details such as resource requirements, project schedule, system, project environment and software requirements.
2. Project Work Breakdown Structure (WBS).
3. Generic software development plans.

Some type of a plan should address each facet of the project. Large plans are not necessary for each task. A task plan may simply be a section of another plan. The important point is that all software development *tasks* and *processes* are addressed based on their complexity, resource requirements, and the degree of risk associated with that task. The items to be addressed by a plan are usually determined by the WBS for the project. The WBS section in this report addresses this concept in more detail.

There are shortcomings created by development plans that are prepared *without* adequate depth and breadth relative to the complexity or magnitude of the project. Any of these factors will contribute to software development project failures, such as cost overruns, inadequate capability, missed schedules, and customer dissatisfaction. A software project should have one master schedule, and all subordinate schedules should be integrated and consistent with the master schedule. The plan should then be approved by management prior to committing major resources to the project. The plan should be frequently reviewed and updated as conditions change during the life of the project. Frequent reviews will ensure that the project is progressing on the correct course delineated by the current plan.

Figure 1: Project Planning Steps

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*Resources, Schedule, Requirements, etc.

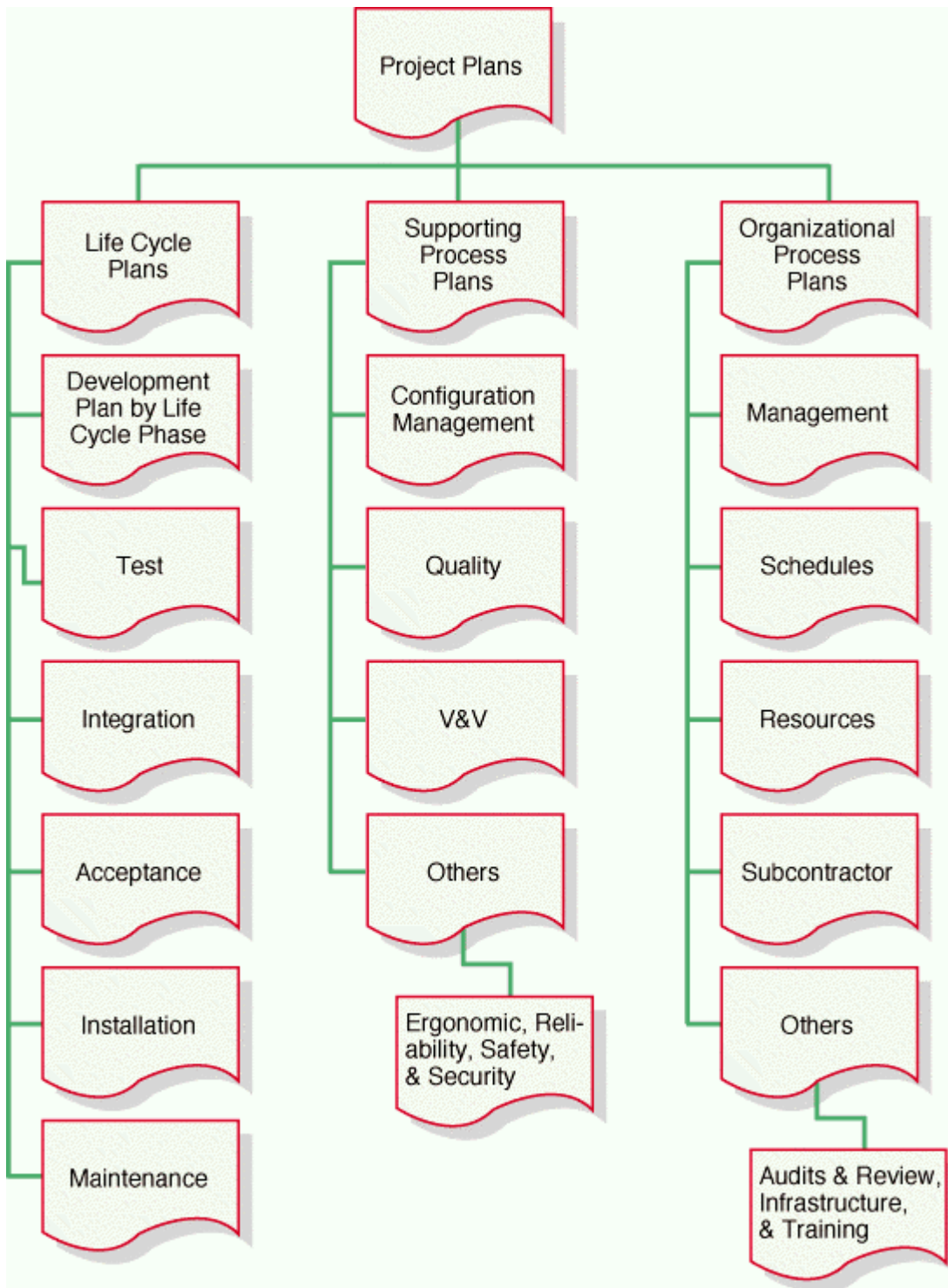
Detailed Planning Steps

The detailed planning process has several steps. The first step is to determine what facets of the development projects need to be addressed by an individual plan, and which may be addressed by a section of another plan. The project should start out with the project plans that were used on the last project and a list of the most common software tasks in order to determine if that process or task is included in the project WBS. If a software-engineering topic such as system reliability is *not* required for a project, this decision and the rationale for the decision should be documented in the project records. This will provide an audit trail and ensure that rational thinking went into the decision *not* to prepare a plan. If a plan is deemed necessary, the next step is to determine the complexity and the magnitude of *each plan* in relationship to the total project and whether a separate plan or just a task within a plan is required for this activity or task. These decisions are predicated on the complexity of the task, risk involved, and the resources required to accomplish the process. This can be shown pictorially as the Planning Document Tree.

Figure 2: Suggested Planning Document Tree

This suggested document tree is collapsed or expanded depending upon the needs of the project.

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Most Common Tasks Requiring Plans

The Tables "*Life Cycle Topics*," "*Supporting Process Topics*," and "*Organizational Process Topics*" define the names of the most common tasks or processes that require plans, with suggested sections for each plan. The life cycle topics deal with the various phases of the life cycle; the topics could be sections of other plans depending on the complexity and magnitude of the task.

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Table 1: Life Cycle Topics

Sections of Plan (alphabetical order)	Requirements Definition	Design (Architectural & Detail)	Test	Integration	Acceptance	Installation
Approach	x	x	x	x	x	x
Approval Matrix for Key Documents	x	x	x	x	x	x
Associated Risk	x	x	x	x	x	x
Constraining Factors	x	x	x	x	x	x
Conversion		x	x	x	x	x
Definition and Background	x	x	x	x	x	x
Definition of Software Changes			x	x	x	x
Deliverables	x	x	x	x	x	x
Disposition of Open Items	x	x	x	x	x	x
Entry Criteria	x	x	x	x	x	x
Exit Criteria	x	x	x	x	x	x
Hardware, Software, and Networks to Be Installed						x
Inputs	x	x	x	x	x	x
Life Cycle Models	x	x				
Outputs	x	x	x	x	x	x
Performance Issues	x	x			x	x
Planned Reviews and Audits	x	x	x	x	x	x
Progress Measures	x	x	x	x	x	x
Purpose	x	x	x	x	x	x
Quality Metrics (How Success Is Judged)	x	x	x	x	x	x
Resources	x	x	x	x	x	x

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Sections of Plan (alphabetical order)	Requirements Definition	Design (Architectural & Detail)	Test	Integration	Acceptance	Installation
Responsibility Matrix	x	x	x	x	x	x
Schedule	x	x	x	x	x	x
Scope	x	x	x	x	x	x
Standards and Procedures	x	x	x	x	x	x
Test Scripts			x	x	x	x
Tools and Techniques	x	x	x	x	x	x
Traceability Matrix	x				x	x
Unique Sections (As Required)	x	x	x	x	x	x

The Table "Supporting Process Topics" shows the supporting processes such as configuration management, quality, and verification and validation (V&V). As in the previous table, these topics may be a section or sections of another plan depending on the complexity and magnitude of the task.

Table 2: Supporting Process Topics

Sections of Plan	Conf. Management	Documentation	Quality	V&V	Ergonomics	Reliability	Safety	Security
Purpose	x	x	x	x	x	x	x	x
Scope	x	x	x	x	x	x	x	x
Approach	x	x	x	x	x	x	x	x
Approval Matrix for Key Documents	x	x	x	x	x	x	x	x
Associated Risk	x	x	x	x	x	x	x	x
Constraining Factors	x	x	x	x	x	x	x	x
Definition and Background	x	x	x	x	x	x	x	x
Deliverables	x	x	x	x	x	x	x	x

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Sections of Plan	Conf. Management	Documentation	Quality	V&V	Ergonomics	Reliability	Safety	Security
Inputs	x	x	x	x	x	x	x	x
Outputs	x	x	x	x	x	x	x	x
Planned Reviews and Audits	x	x	x	x	x	x	x	x
Quality Metrics (How Success Is Judged)	x	x	x	x	x	x	x	x
Resources	x	x	x	x	x	x	x	x
Responsibility Matrix	x	x	x	x	x	x	x	x
Schedule	x	x	x	x	x	x	x	x
Standards and Procedures	x	x	x	x	x	x	x	x
Tools and Techniques	x	x	x	x	x	x	x	x
Unique Sections								
Ergonomics Requirements					x			
Hazard Analysis							x	
Privacy and Security Requirements								x
Reliability Requirements						x		

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Sections of Plan	Conf. Management	Documentation	Quality	V&V	Ergonomics	Reliability	Safety	Security
Safety Requirements						x		
Traceability Matrix					x	x	x	x

The Table "*Organizational Process Topics*" deals with management plans, resource plans and schedules, and all other activities that affect all segments of the project organization. The organization may include other topics for software development plans if the requirements of the project so dictate. As with the life cycle and supporting process topics, the organizational process topics may be a section or sections of another plan depending on the complexity and magnitude of the task.

Table 3: Organizational Process Topics

Sections of Plan	Management	Schedules	Resource	Audits and Reviews	Infrastructure Support	Sub-Contractor	Training
Purpose	x	x	x	x	x	x	x
Scope	x	x	x	x	x	x	x
Definition and Background	x	x	x	x	x	x	x
Approach	x	x	x	x	x	x	x
Approval Matrix for Key Documents	x	x	x	x	x	x	x
Constrainting Factors	x	x	x	x	x	x	x
Deliverables		x	x	x	x	x	x
Inputs		x	x	x	x	x	x
Outputs		x	x	x	x	x	x
Planned Reviews and Audits	x	x	x		x	x	x
Policies and Procedures	x	x	x	x	x	x	x
Risk	x	x	x		x	x	x
Quality Metrics	x	x	x	x	x	x	x

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Sections of Plan	Management	Schedules	Resource	Audits and Reviews	Infrastructure Support	Sub-Contractor	Training
(How Success Is Judged)							
Resources	x	x	x	x	x	x	x
Responsibility Matrix	x	x	x	x	x	x	x
Schedule	x	x	x	x	x	x	x
Standards and Procedures	x	x	x	x	x	x	x
Tools and Techniques	x	x	x	x	x	x	x
Unique Sections							
Administrative Function	x						
Agendas for Reviews				x			
Change Control		x				x	
Class Schedule							x
Completion Criteria				x		x	
Courses Requirements							x
Customer Furnished Resources			x				
Handling of Open Items						x	
Life Cycle Cost			x				
Management Reserve		x	x				
Organization Chart	x						
Personnel Acquisition			x				

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Sections of Plan	Management	Schedules	Resource	Audits and Reviews	Infrastructure Support	Sub-Contractor	Training
Plan							
Required Infrastructure Support					x		
Return On Investment (ROI)			x				
Skill Progress Path							x
Work Breakdown Structure		x	x				

Preparing the Plan

There are ten steps to plan preparation:

1. Create a base of knowledge. This is accomplished by gathering information such as old similar plans, lessons learned, related software engineering books, standards, and company procedures.
2. Prepare an annotated outline of the plan. This outline should include page and figure counts as well as the person responsible for each section.
3. Generate a schedule and budget estimate for preparing the plan. For most projects, there is a project directive that will give each individual plan's overall budget and schedule parameters for this step. The schedule and budget estimation should be synchronized with the other project plans.
4. Compile a list of people who should review the completed plan. This committee should contain appropriate technical experts who have had experience on several development projects.
5. Obtain approval of the annotated outline, budget, schedule, and review committee. This step will decrease the amount of rework that will be necessary.
6. Prepare a first draft of the plan. This draft should contain the schedule for reviews and updates of the plan.
7. Coordinate the data in the plan with all other affected plans. This step will reduce conflict among project personnel.
8. Conduct a review of the plan with the review committee.
9. Incorporate changes.
10. Obtain buy-off approval of the plan. When approval has been received, the plan should be placed under change control.

Planning Costs

The cost of a plan will run between 8 to 16 labor hours per page. The minimum flow time to put together a plan of more than 3 pages is 40 hours. This is due to the amount of coordination required to ensure that

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all project plans and all aspects of the project dependencies relating to each other are identified and synchronized. Adequate plans will ensure the success of the project and are well worth their cost.

Questions the Plan Should Address

- What is the task or process?
- Why is the task or process being done?
- Who will do it?
- When will it be done?
- How will it be done?
- What are the quality criteria for this task or process?
- What are the risks?
- What are the inputs?
- What are the outputs?
- What are the constraining factors?
- What are the deliverables?
- Who will approve all documents associated with this task or process?
- When and how is the process or task completed?
- How are open items handled?
- What standards and procedures are to be used for the process or task?
- How is success determined?
- What are the resources required?
- What reviews and audits are planned?

Common Mistakes in the Software Planning Process

- Plans do not convey the high-level requirements or risk to senior management.
- Configuration management plan is not detailed sufficiently to manage change.
- Estimating cost model does not address the volume of software to be produced or is not calibrated to the individual project.
- Only one cost model is used.
- The handling of interfaces between software, hardware, and people are not planned in enough detail.
- Plans do not adequately address the exit criteria for the task or process.
- Plans do not address the quality requirements for the software or documentation products.
- Plans do not address ownership rights, such as warranties, intellectual rights, patents, and copyrights of the software.

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The Review Process

The purpose of a review is to ensure that the project is following the plan or meeting the specifications according to the plan. Properly executed and timed reviews also ensure that problems are detected early on when corrective action can be taken without incurring additional costs or missed schedules. Serious problems occur when appropriate reviews are not conducted or the wrong issues are reviewed.

There are five types of recognized software development reviews:

1. Management.
2. Technical.
3. Inspections.
4. Walk-throughs.
5. Audits.

Reviews may be formal or informal in nature. (This report will only address management reviews and technical reviews that are of a formal nature.) Formal reviews are those that meet distinctive attributes, while informal reviews are those that do not meet the requirements for the formal review process. A formal review as defined by ANSI/IEEE-STD. 610.12-1990: *Standard Glossary of Software Engineering Terminology* is a meeting during which a software product(s) or plan(s) is presented to qualified and interested parties for review, comment, and or approval. A formal review will have a defined objective, scope, process, and procedure. Formal reviews may be conducted in a designated meeting place with all parties in attendance; via telephone, video, or other electronic media; or a combination of media. A formal review has the following attributes:

- Team participation.
- Documented procedure for conducting the review.
- Written reports.
- Accountability of the review team for the quality of the review.
- Measurement of process effectiveness.

In addition, the formal review procedure defines:

- The purpose and objective of the review.
- The roles and responsibilities of the participants.
- The input required for the review.
- The entry criteria for the review.
- The exit criteria.
- The outputs of the review.
- The evidence that is required to support an audit of the process.

Management reviews are periodic in nature and are not tied to a specific milestone. A management review is an evaluation of a software development process performed by, or on behalf of, senior management based on project status relative to a plan. All major elements of the plan are reviewed.

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These reviews identify deviation from the plan and provide a forum for recommendation on how to correct the situation with follow-up action plans and due dates. Correction can occur by either correcting the deficiency or modifying the plan.

Specification or technical reviews are usually conducted at major milestone points of the product life cycle, such as requirements definition, specification development, design, and acceptance. A technical review is an evaluation of the quality of the software products that is performed by a team of qualified experts. Technical reviews identify any discrepancy from specifications or standards and provide a forum for the examination of alternatives and recommendations on how to correct a situation with follow-up action plans and due dates. Correction can occur either by fixing the product or by modifying the standard or specification.

A new approach being tried by some progressive companies is called "ongoing technical oversight." Using this approach, a senior technical person is responsible for tracking the technical performance of the project on an ongoing basis to determine if the project is meeting its technical goals. This approach is successful only if the person responsible for the oversight role is well respected by the project team and has access to key individuals and documents.

A software development project may have several reviews associated with it. Some of these may occur more than once depending on the magnitude and complexity of the project. The following tables contain suggested lists of the most common reviews associated with a development project. Project managers may find that other reviews are required, and they should be added to this list. Other reviews may be combined or deleted depending upon the complexity and magnitude of the project.

Table 4: Management Reviews— Progress

Review	Purpose	Frequency	Review Items
Progress	Review the progress of the project and take corrective action as required	Weekly to monthly, depending upon the size and complexity of the project	• Action items • Budgets • Quality indicators • Risk abatement plans • Schedules

Table 5: Management Reviews— Plans

Reviews	Purpose	Frequency	Review Items
Plans as listed under "Review Items" column. Reviews for each plan should be held separately	Ensure that the plans are being followed as documented	Plans should be reviewed at least every 1 to 3 months during a project life cycle	Project plans as listed below: • Audits and Reviews • Configuration Management • Documentation • Ergonomics • Infrastructure Support • Installation • Maintenance • Quality • Reliability • Safety • Security • Test • Training

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Reviews	Purpose	Frequency	Review Items
			• V&V

Table 6: Technical Reviews (Typical)

Review	Purpose	Frequency	Review Items
Project Go Ahead	Ensure that the project plans are adequate and they address all facets of the project	Completion of planning phase	Project Plans
Requirements Definition	Ensure that the project requirements are clear and concise and address the customers' or marketing needs	Completion of requirements phase	Requirements Document
Software Architectural Design	Ensure that the software design meets the requirements specification and is technically achievable	Completion of software architectural design	Architectural Specification
Detail Software Design	Ensure that the module design meets the software specification and is technically achievable	Completion of software design	Software Module Specification
Coding	Ensure that the software design has been converted to code correctly. (This can be achieved by a structured	Completion of module coding	Program Listing

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Review	Purpose	Frequency	Review Items
	walk-through instead of a formal review.)		
Unit Testing	Ensure that the modules have been adequately tested	Completion of unit testing	Test Records and Open Items
Integration and System Testing	Ensure that the system has been adequately tested	Completion of integration testing	Test Records and Open Items
Acceptance	Ensure that the system and all products meet the requirements specifications	Completion of the system and all associated products	Requirements Specification, Test Records, V&V Records, and Open Items
Delivery, Installation and Training	Ensure that site installation and training has met the customer requirements	Completion of a site installation and training	Customer Requirements Specification, Test Records, V&V Records, Open Items, and Training Requirements and Records

Key Participants

Two key persons—the review leader and the recorder—are essential for a review to be successful. The review leader is responsible for the planning and preparation of the review and all administrative tasks associated with the review. The leader has the responsibility to conduct the review in an orderly manner and ensure that the review meets its objectives. The leader is also responsible for issuing the review report. The leader must ensure that the group focuses on the issues and not on a person or organization. The recorder is responsible for documenting all significant deviation from plan, specification, or standard. The recorder records all decisions and recommended changes. The review results should be documented in such a way that auditability and traceability are achieved.

Review Steps

Five review steps are necessary to conduct any formal review:

1. *Authorization.* The need for a formal review and its authorization are covered in the review plan. In addition, a formal review may be called for by management or by the customer at any time, depending upon the circumstances.
2. *Planning.* The planning activities include identifying the appropriate management support, identifying the participants, preparing the agenda, scheduling the meeting place, and distributing the material to be reviewed. The review leader is responsible for the planning activities.

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3. *Preparation.* Each participant should review the relevant section of material well before the meeting and prepare a listing of potential anomalies by classification (major and minor). This listing should be discussed in advance with the review leader, if time permits, in order to expedite the review and prevent possible disruption to the process.
4. *Conduct the Review.* The review should identify the true major anomalies from the plan or specification. Following appropriate input from the participants, a recommended course of action should be determined for each anomaly. If other issues surface that are not covered in the initial agenda, a method for addressing these issues should be determined. It is important that the review remain focused and not become sidetracked by irrelevant issues. If the review is not completed to the satisfaction of the review leader, subsequent meetings should be scheduled.
5. *Follow-up.* A report that documents the proceedings of the review should be generated in a timely manner. This document should list the anomalies, issues, and recommended actions. The review leader should ascertain that the actions decided from the meeting are resolved.

Review Costs

The costs for reviews are high. This is due to the number of people necessary to conduct an adequate review and the labor hours that are associated with the planning and follow-up action. However, if the right problems are identified and resolved in a timely manner, reviews are invaluable, and totally warrant the expenditure.

Common Mistakes in the Software Review Process

- Key questions are not asked and answered.
- Not assuring that all of the potential problem areas are addressed and documented.
- The review focuses on personality issues rather than on the substantive issues affecting the project.
- Use of a team leader that is not skilled in conducting reviews.
- Not having the appropriately knowledgeable people in attendance.
- Inadequate preparation time.
- The review team is not adequately informed about the issues prior to the review.
- Extra or unnecessary people in attendance.

Work Breakdown Structure (WBS)

According to the Project Management Institute, a work breakdown structure (WBS) is a *product*-oriented "family tree" that allows the total scope of the project to be organized, scheduled, and defined. The WBS incorporates both the *product* definition and the *process* to build it; these two elements compose the WBS. The top level of the WBS is the product. Each descending level provides a more detailed description of the product and its associated process. The higher levels of a WBS incorporate the more general elements, while at the lower levels the elements become more specific.

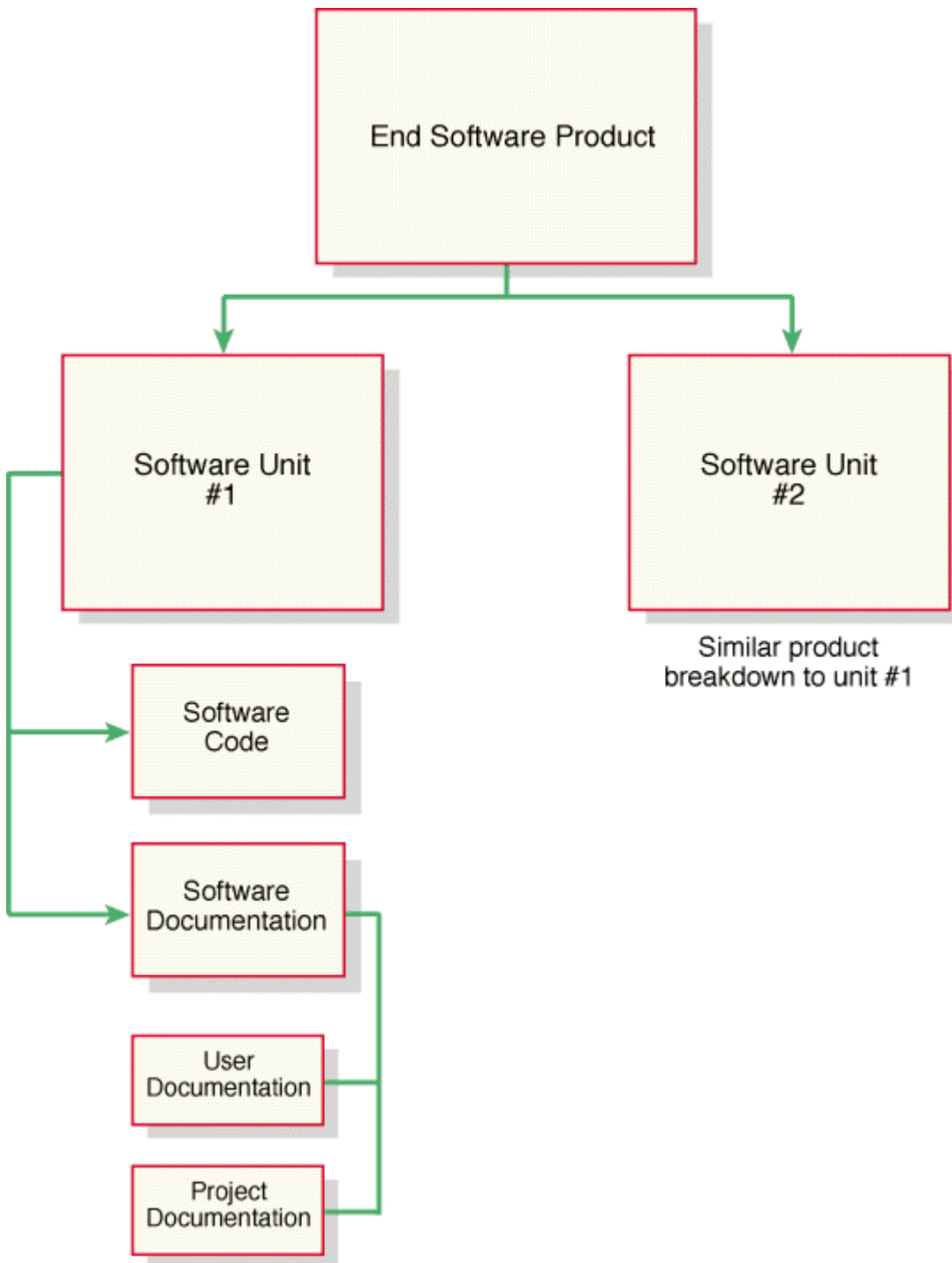
Product WBS Element

The product WBS element shows a system that may subdivide a product into all of its physical components. Associated with each component is the task output required to produce a component such as an engineering drawing or specification. In developing software, this concept is elusive since the end product is usually a disk or tape, and the intermittent components are software units that may or may not

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have physical representation except as data stored on a disk. These physical outputs can be visualized hierarchically as a tree in which the end product is at the top of the tree, and each component with its associated task output is a branch of this tree. An example of an associated task output would be the requirements specification for a particular software unit.

Figure 3: Software Product WBS Example



Process WBS Element

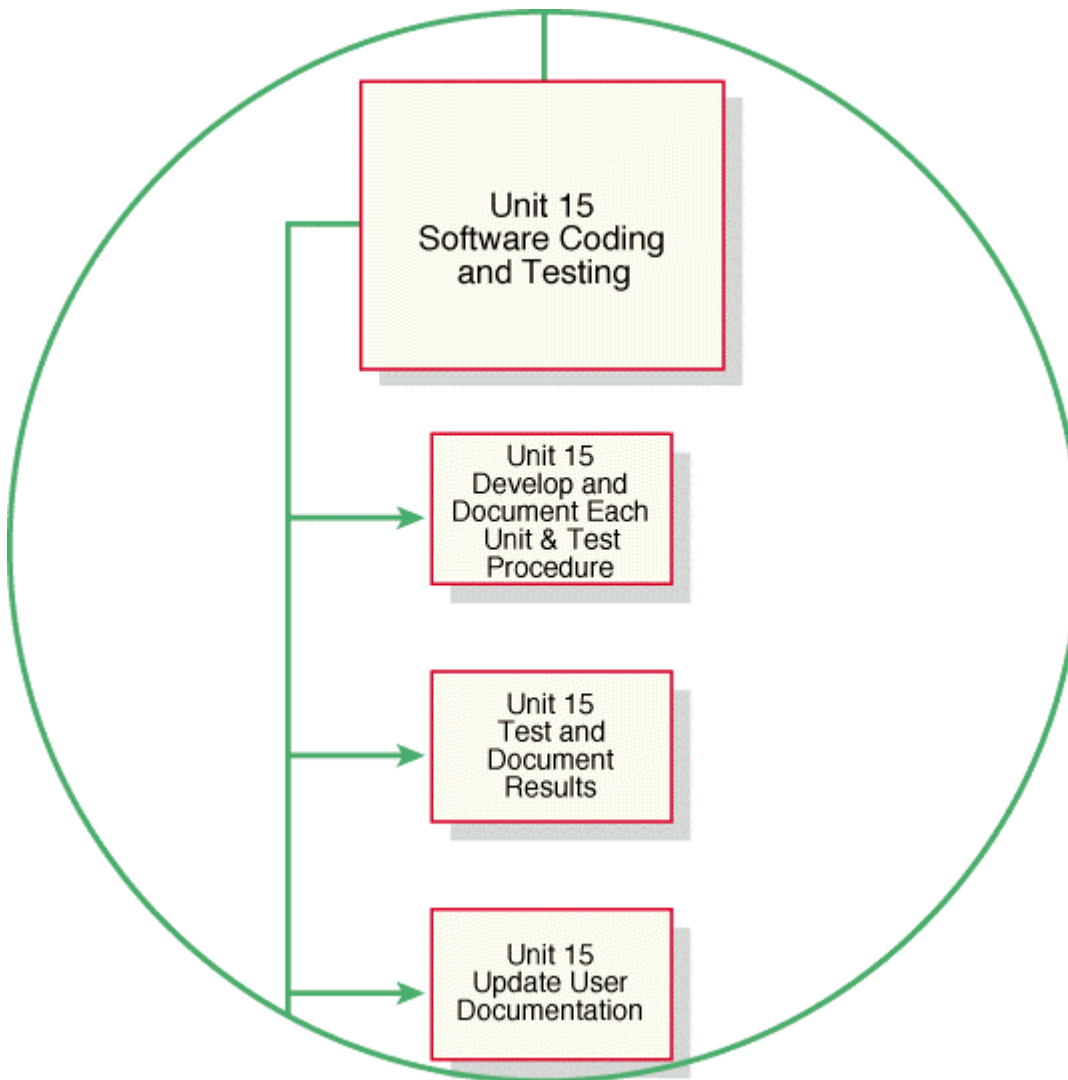
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A system that logically divides a process into lower activities and tasks is the Process WBS Element. In software projects, the process of development would be subdivided into the activities of requirements definition, software design, software construction, and software testing. The activity, software design, would be subdivided into software architectural design and software detail design.

Product and Process WBS

The Product and Process WBS is a combination of these two work breakdown structures. This combination produces a WBS number that represents such tasks as coding and testing a particular software unit.

Figure 4: Process and Product WBS Example



Conclusion

Many software project managers do not adequately apply planning and review processes when managing their projects. To have a successful software development project, adequate plans must be developed in the first phase of a project. These software development plans must consider all aspects of the project and all perspectives. Plans should cover subject matter from pure management issues such as cost and schedules, to the technical issues of the project such as testing, security, and performance. These plans

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must be created from a knowledge base that is credible and proven, prepared in a systematic way with annotated outlines, and reviewed and approved by management. These plans become the baseline of a project and the cornerstone for the review process.

The purpose of reviews is to ensure that the project is following the plans and building products to a predefined specification and to detect problems at the earliest possible time, so that corrective action may be taken. The project review must be well defined at the project level and adequately planned, staffed, and executed. Action items generated by the review must be pursued and resolved. Technical reviews should be the vehicle for closing major technical milestones. Management reviews should occur periodically to ensure that the project is progressing satisfactorily in meeting cost, schedule, and quality targets.

A software development project with adequate plans and timely reviews will substantially increase its odds for project success by meeting cost, schedule, and technical commitments. When well executed, these two important management processes— planning and reviews— increase the odds for a software project's success.

This report was prepared exclusively for Gartner Group's Datapro by Stan Magee of Software Engineering Process Technology (Issaquah, Washington). Mr. Magee is acting convener of WG 7 (Software Life Cycle for ISO SC 7) standards group. Mr. Magee has over 35 years experience in the software field and is considered an expert in the area of software life cycle methodology. He specializes in software engineering process technology, serving both national and international clients.

References

ANSI/IEEE-STD 610.12-1997: *Standard Glossary of Software Engineering Terminology*. American National Standards Institute/Institute of Electrical and Electronics Engineers.

ANSI/IEEE-STD 1028-1997: *Software Reviews and Audits*. American National Standards Institute/Institute of Electrical and Electronics Engineers.

ISO 9000-3-STD-1997: *Quality Management and Quality Assurance Standards— Part 3, Guidelines for the Application of ISO 9001 to the Development, Supply, and Maintenance of Software*. International Organization for Standardization.

ISO/IEC 12207 STD-1995: *Information Technology-Software Life Cycle Processes*. International Organization for Standardization.

ISO/IEC TR 16326-1998: *Software Engineering-Project Management*. International Organization for Standardization.

Humphrey, Watts S. *Managing the Software Process*. Software Engineering Institute. Addison-Wesley Publishing Co.. 1990.